

Lec 45 Mathematics of Diffusion Models

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Transcript

[00:00] Hello, everyone. Welcome back to the lecture series on fundamentals of generative AI and large language models. In my previous session, we have discussed about diffusion models. I have given you some idea of how diffusion models, what exactly is diffusion model, and how it works in a very, you know, high-level way, okay? So, to understand diffusion model much more in-depth, there are some mathematical concepts that are required because each and every step we go ahead with the diffusion model actual working, we will require these fundamentals of math maths.

[00:42] And today, in my today's discussion, I will discuss about the mathematical foundations of diffusion model, okay? So, the first thing that we will do here today is we will set up the problem statement. So, we will see what happens in diffusion models, what are the processes involved, and then we will see what is the math involved or what are the foundational concepts of math that is involved to understand the entire mathematical formulation of diffusion models, okay?

[01:14] So, let us start with setting up the problem statement. We all know that in diffusion models, we should diffusion probabilistic models, there are two main processes. The first one is the forward diffusion process, in which we take an a clear

image, and then we keep on adding noise to that one till it goes to almost pure noise state, okay? So, we have a picture here of a dog jumping. And then, if you see, this is at t equal to zero, then at t equal to one, there is a little bit of noise added to it. At t equal to 2, there is more noise added and in the subsequent steps, you know, gradually increasing amount of noise is getting added. And the final state at t equal to capital T , we get almost pure

[02:11] noise. Okay? So, this is the forward diffusion process and we are aware of that one, right? We have discussed about this one in the last session also. Now, here x_0 is the original data, which is your image or the signal. Here it is the clean image. What is x_t ? x_t is the noisy version at time step t . And capital T is the total number of diffusion steps. Okay? And capital T can go from 1 to 1,000 also. Okay? So, depending on the application, the number of steps will be chosen.

[02:50] And small t belongs to a set from 1 to capital T . Okay? So, this is about forward diffusion process. Now, we have we usually write it in the terms of probability distribution, as we always do. So, this is written as Q within bracket x_t given x_{t-1} . » [snorts] » What exactly is meant by that one? It means the probability distribution of x_t conditioned on x_{t-1} . So, if I know x_{t-1} , suppose I know I I uh know x_2 , then my x_3 will be conditioned on x_2 .

[03:37] Okay? So, this is how we write the probability distribution for forward diffusion process. And in the forward diffusion process, it's a fixed process. It's a known process. Why it is a fixed process or known process? Because we are not learning anything here. We are taking a a clear image and then step-by-step we are adding noise so that it goes to a completely you know, pure noise state. So, in forward diffusion basically what we are doing, we are generating a noisier sample from a cleaner sample.

[04:13] And since the forward diffusion process is known and fixed, this distribution can be pre-defined as a Gaussian distribution, okay? So, we are not learning anything in the forward diffusion process. We are considering it as a pre-defined Gaussian distribution, okay? So, this is about the problem statement that we have in the forward diffusion process, okay? Now, let us look into what is the reverse diffusion process.

[04:45] In reverse diffusion process, what we do is we start from T equal to capital T , which is the almost pure noise state. And then each and every state, we denoise that image. And we keep on doing it, iterating it over multiple number of steps till we get the clean image which was our x_0 . Okay? So, this is the reverse diffusion process. And in the reverse diffusion process, that there that is the place where we get the learning. The neural network learns.

[05:22] What it learns? It learns to predict how much of noise have been put in one particular time stamp time step T , okay? So, that is what it will try to predict. So, there is a learning of the neural network in the reverse diffusion process, okay? And

this is the probability distribution. We write it like this, $P(\theta_{x_t-1} | x_t)$. Now, what exactly I uh you know, what does this probability distribution equation means? This probability distribution of x_{t-1} is conditioned on x_t . Okay? Because we are coming in the reverse direction. So, I will first get x_{t-1} .

[06:09] And that will be conditioned on x_t . Okay? So, we are coming from t equal to capital T , then we come to a place where t equal to small t , and then we come to a place where x you know, t equal to $t-1$. So, and then finally we get the clean image, you know, at t equal to 0. Okay? So, this is just in the reverse direction we are doing. So, what happens is we are starting from here. This I have taken because I want to show the reverse direction. Okay? t equal to capital T . So, in the reverse direction what is happening? You your x_{t-1} is conditioned on x_t .

[06:52] And if you see, there is a term θ here, which was not there in the forward diffusion process. Okay? So, what is this θ you know, what exactly it means? What what what what information θ is giving? θ is basically these are the parameters of the neural network. Okay? So, θ refers to all the trainable weights of the denoising neural network. Fine? And in reverse diffusion, we generate cleaner sample from a noisier sample.

[07:30] In the reverse diffusion, what we do? We start from almost pure noise, and we come to the cleaner one. Okay? So, that is the problem statement of reverse diffusion process. Now, we have seen that noise is added in the forward process, and noise is taken out of the images in the reverse process. So, noise is a very important factor in case of diffusion probabilistic models. Now, we will learn a new term which related to noise, which is called as the noise variance schedule. Now, what exactly is noise variance schedule?

[08:10] Noise variance schedule is basically a sequence of values, like $\beta_1, \beta_2, \dots, \beta_T$. It controls how much of noise have been added in each diffusion step. Okay? In the forward diffusion step, what happens is we keep on adding noise, but the amount of noise that is added in each and every step is different. Initially, when T equal is small, the amount of noise added is small. When as and when we progress for higher values of T , then amount of noise added is higher, okay? So, this is called as the noise variance, okay?

[08:54] So, β_T actually gives you the noise variance schedule, which is telling that it is the noise added is not constant, it changes with time, and β_T belongs to you know, a number between zero and one. And β controls the rate at which the noise is injected, and the signal-to-noise ratio decreases, okay? So, it controls that β value controls the noise, how much noise is injected, and when noise more noise is injected, the signal-to-noise ratio it changes, okay? So, signal-to-noise ratio decreases, and so it β will decide how much will be the signal-to-noise ratio at different time steps.

[09:36] For smaller T , the noise added is small and for larger t , bigger amount or larger amount of noise is added. And this gradual increase in the noise addition is called as the noise schedule. Okay? Now, what is the problem if I add same amount of noise in all the steps of diffusion? Why we require a schedule? Is Why Why it is important? So, we'll have a look into that one. Now, in the forward diffusion process where noise is added, we start from the clean image and we want a complete destruction of the image at my last step, t equal to capital T .

[10:19] Okay? So, information destruction rate or the complete destruction of the information that is present in the image, initial image, that I want at the last step of my um noise addition process. Now, if the information destruction rate is too fast or too low, there are some challenges. We will see what happens when the destruction uh information destruction rate is too fast. In that case, the data becomes pure noise too quickly. Okay? That means it is very It is just, you know, from maybe in three, four steps or five to six steps, if we change a clean image to a pure noise, the learning in the reverse process becomes difficult because the changes that have happened to the data within a very small

[11:12] amount of time step. If it is If If the learning rate is Oh, sorry. If the uh you know, way the noise is added, if it is too fast, then the learning of the amount of noise added in each and every step becomes difficult. Okay? Whereas, if it is too slow, there also there are challenges. There will be too many steps involved between the clean image and the completely noise image. Okay? Pure noise. And in that case, the training will will be inefficient and too many steps will be required.

[11:50] So, that is the reason a schedule is important, which will decide how much of noise is getting added in each and every diffusion step. Okay? So, as I was telling, we want X_T , that is the image at time instant capital T , to be completely noise, like $N(0,1)$. It's a Gaussian noise. Okay? With mean zero and variance one. And the scheduler must a schedule must ensure complete loss of structure by steps T . So, that is very important.

[12:27] Within that number of steps, the information should get completely uh you know, removed from that image. Okay? Now, as as we keep on moving forward in the diffusion probabilistic model, uh you know, understanding the diffusion probabilistic model, we will be talking about Gaussian distributions almost everywhere. Okay? So, let us have a quick review of what is this Gaussian distribution, what is standard Gaussian, and what are the things that are associated with the Gaussian distribution. Okay? So, let us have a look into what is Gaussian distribution. So, basically, Gaussian distribution, it's a probability distribution that describes a data that is clustered around the mean.

[13:18] It is it is clustered around the mean in symmetric both in the right side and left side, in both the sides. Symmetric spread. Okay? And the probability distribution

is given by this equation $\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x-\mu}{\sigma}\right)^2$. Where x is the random variable which is your basically your data point. μ is the mean. σ^2 is your variance or the spread in the data.

[13:49] And σ is your standard deviation. Okay? So a quick look into a Gaussian distribution. So this is my mean and this is $\mu + \sigma$ $\mu - \sigma$ and the, you know, number of points, you know, 68% of the data, you know, is here between $\mu + \sigma$ and $\mu - \sigma$. And then $\mu + 2\sigma$ and $\mu - 2\sigma$ between those two ranges we have 95% of our data and $\mu + 3\sigma$ $\mu - 3\sigma$ we have 99.7% of the data. Okay? So this is basically a Gaussian distribution and this factor is basically a normalization factor which ensures the total probability equal to one. Okay? And this exponential term it tells that the probability density decreases exponentially as the distance from the mean increases.

[14:52] So you can see this is the mean. And as the distance between the mean uh distance of the data from the mean keeps on increasing, the probability distribution you know, uh goes down. Okay? So that is that is what is ensured by this exponential term. Okay? Now we will see what is a standard Gaussian. Okay? So now we know what is a what is a Gaussian distribution. It is centered at mean and then we have standard deviation σ , okay? And variance is σ^2 .

[15:32] Now, when μ equal to zero and the variance is one, then we call it write it as $N(0, 1)$. This is the mean and this is the variance, okay? And we call this type of Gaussian as the standard Gaussian. When the mean is zero and variance is one, we call it as standard Gaussian. Okay? Now, we will have a look into another new term which is multivariate Gaussian. Now, what exactly we mean by multivariate Gaussian? From the name itself, you can see multivariate. That means it is related to something which are having more than one variable.

[16:12] So, a multivariate Gaussian distribution is a generalization of Gaussian distribution with multiple variables, okay? Which is basically in a form of a vector. So, here what happens is if you have in our previous slide, we have seen in Gaussian distribution, we took the data point as x . Now, here instead of a single number x , we have x belonging to a d -dimensional space, okay? So, it has it is in the d -dimensional space means it has d number of you know, components.

[16:52] And if you see the probability density function modeled as a multivariate normal distribution with mean vector μ and covariance matrix σ is written like this. So, this is basically a probability uh distribution or probability density function and this is written as x which is your uh vector, multiple variable which is a vector. And this is your mean μ . And here instead of variance σ^2 , there is a new term that has a new symbol that has come up, which is called as the covariance matrix. Okay?

[17:31] So, whenever we talk about multivariate Gaussian, we will not speak about variance, but we will speak about covariance. Okay? Now, the question arises, why we require multivariate Gaussian? Is it required in our, you know, whatever topic of discussion we are having it today? What is the relevance of multivariate Gaussian with our this diffusion probabilistic model? Let us have a look. So, in machine learning or deep learning or in generally VI, we rarely use a single variable.

[18:12] It is always a vector. So, like image will have millions of pixels. Okay? In any embedding, it is in the high high dimensional vector. Okay? So, that means we don't speak about a single variable. We speak in term data, we speak of data in in higher dimensions. And whenever there are, you know, data in higher dimension, we require multivariate Gaussian distributions. Okay? So, if X my X vector belongs to a D dimensional space, I can write it like this as a vector.

[18:48] And » [clears throat] » there are dependencies between these, you know, com- these X_1, X_2, X_3, X_T . So, pixels in an image are correlated. Features in a data set are correlated. Okay? So, relationship between the features is captured by covariance. Okay? So, that is the reason we require the concept of covariance to come into picture while we are speaking about diffusion probabilistic models. Okay? Now, we'll have a look what exactly this covariance I told that it captures the relationship between two variables. Let us have a quick look into what I mean by that one.

[19:34] So, covariance measures how two variables change together. Okay? Here I have two pictures. One is ice cream sale in hot summer and another one is hot chocolate sale in hot summer. So, we have two features say in a data set. One is the temperature of a place and the weather temperature and the sale of ice cream. Okay? Now, we want to see is there any relationship between the temperature of that place with the amount of ice cream that is sold.

[20:11] If we see as the temperature increases as summer, you know, uh becomes intense, the number of ice creams that is getting sold is increasing. So, this is basically an example of positive covariance. There is a relationship between the temperature weather temperature at that place and the amount of ice creams that are getting sold. Okay? So, with increase in temperature the amount of ice creams that are getting sold also is increasing. So, this is an example of positive covariance. Whereas, if I have weather temperature as one feature and hot chocolate sale another feature in summer.

[20:55] So, we will see as the summer becomes intense, the amount of hot chocolate sale may not increase that much. It may reduce. So, this this is called as or this is an example of negative covariance, okay? And we spoke about covariance matrix when we were speaking about multivariate Gaussian. We will look into how the covariance matrix looks like. So, the covariance matrix basically looks like this. The diagonal ele-

ments, if there are two uh vari- you know, variables X_1 X_2 , then the diagonal elements will give the variance of X_1 and variance of X_2 .

[21:39] And the off-diagonal elements gives covariance of X_1 X_2 , this one. And the bottom is covariance of X_2 X_1 , okay? So, this is how the covariance matrix is calculated, okay? Now, let us have a quick look into how to uh calculate the covariance matrix. We like to have a numerical example so that it becomes clear to you all. Suppose we have a data set with height and weight of human being, okay? And so, for a person, there are two features, one is height and one is weight, okay? So, X_1 is your height, X_2 is your weight.

[22:27] So, covariance matrix, I have kept the formula here in front of me. So, from my previous slide, so it looks like this. If I want to make this matrix, I need to calculate the variance of X , covariance of X X_1 X_2 , variance of X_2 , covariance of X_2 X_1 . So, in this example, we will see how to do that one. So, this is my data set, person A B C D, and they have two features, height and weight, and these are the values, 150 50, 160 for person B, 160 55 person C, 170 65. For person D, 180, 72. The heights and the weights, okay? Height is X_1 and weight is X_2 .

[23:12] Then what? We will find out the mean height, which is which we will call as μ_1 . So, if you calculate, you will see it is 165. And the mean weight calculated is 60.5. Okay? So, we have μ_1 and μ_2 . Okay, which is the mean height and the mean weight. Then what do we do? For finding out variance of X_1 , it is basically the expectation value of X_1 minus μ_1 whole square. Okay? So, here I have kept the table. Person one, X_1 was 150. X_1 minus μ_1 , if you do the calculation, it is minus 15.

[23:52] X_1 minus μ_1 whole square is 225. Similarly, I have done it for B, C, and D. All the four persons I have calculated what is X_1 minus μ_1 and then X_1 minus μ_1 whole square. Okay? And then we have to take the expectation value or the average value. Now, variance of X_1 is given as the addition of all these terms divided by four. Here we are talking about the population variance. That is the reason we are dividing by n , which is the number of data points that we have. If we are taking sample variance, we have to divide by n minus one. Okay? So, like this, we have calculated variance of X_1 , which is equal to 125. Variance of X_2 is left to the participants to calculate by themselves. Okay? Then we will look into

[24:44] how to calculate the covariance between X_1 and X_2 . So, the covariance between X_1 and X_2 is basically the expectation value between X_1 minus μ_1 into X_2 minus μ_2 . So, again, I have kept my μ_1 and μ_2 values here just to have a quick look so that we don't get we don't have to turn back to the previous slide. And we I have the same table here little bit extended. So, person A B C D is here then x_1 which is your you know, one feature then x_1 minus μ_1 x_2 x_2 minus μ_2 and then x_1 minus μ_1 into x_2 minus μ_2 . So, you all can pause here, cross-check the

calculations and if there is any mistake in the you know, numerical values that are put here, please feel free to reach out to me. That may be there if there is any

[25:37] typo or anything or you can you know, correct by yourself also. So, covariance between x_1 and x_2 if we calculate like this, now we have x_1 minus μ_1 into x_2 minus μ_2 and then we take the expectation value, it becomes 95, okay? Uh you know, you can do the calculations by yourself. So, the now we are making our defining [clears throat] our covariance matrix which is variance of x_1 , covariance of x_1 x_2 in the right top corner, then here we have covariance of x_2 x_1 and variance of x_2 , okay? So, if you see covariance of x_1 and x_2 and covariance of x_2 x_1 , they are basically the same thing because here it will be expectation value of x_2 minus μ_2 into x_1 minus μ_1 . So, basically it is the same value and after calculations

[26:32] and putting all those values here, we get the covariance matrix like this. So, this is this numerical example I have taken up just to give you a clear understanding of how covariance matrix is calculated because I will be speaking about the covariance in my next lectures whenever we are speaking about diffusion probabilistic model. It will very much come in the discussions. Okay? So, with this one I come to the end of my today's session.

[27:06] Let me go through once or let us recap once what all we have studied today. So, we have studied we have started with the main idea of today's discussion is to have a review of the mathematical foundations that are required when we go ahead and look into the mathematical formulation of diffusion probabilistic models. So, for that one we started with setting up the problem statement. So, we have two processes, the forward diffusion process and the reverse diffusion process. We have seen the forward diffusion process the probability distribution can be written in the how it can be written in the mathematical form. And the forward process there is nothing to learn. So, it's a fixed and known process and that

[27:55] is the reason we take that probability distribution for the forward process as a predefined Gaussian process. Whereas in the reverse diffusion process, the neural network learns how to denoise the image. Okay? So, that is where the learning happens. And if we see go back to or we have seen that the probability distribution equation in the for the reverse denoise reverse diffusion process, there is a parameter θ there. And what is this θ ? It it actually you know, relates to the weights and the biases which gets trained during the training phase. Okay?

[28:37] And during the forward diffusion process, it is a conditional probability that given you know x_{T-1} , you can find out x_T . Similarly, in the reverse diffusion process also, it is a conditional probability, okay? Now, once we are thorough about the uh you know, the problem statement, uh we went back to have a quick review on the Gaussian distribution. What exactly Gaussian distribution means and how it looks, you know, the curve for Gaussian distribution, the equation for the probability density

function for Gaussian distribution, and uh we have looked into what is a standard Gaussian.

[29:22] So, when we have a Gaussian with mean zero and variance one, we call it a standard Gaussian. And then, we have looked into what is a multivariate Gaussian. So, instead of a single variable, when we speak about a vector with multiple number of variables in that vector, we go from normal Gaussian Gaussian distribution to multivariate Gaussian distribution. And in that in while we define multivariate Gaussian distribution mathematically, we do not talk about the variance, but we talk about the covariance matrix, okay?

[29:59] So, then we try to understand what is this covariance matrix. What exactly the covariance does? So, covariance basically captures the relationship between uh you know, features inside inside your data uh data set. Now, we have seen how what is we have taken two examples of positive covariance and negative covariance. So, when there is uh with the rise of one uh feature, another feature also rises, that we call it positive covariance.

[30:30] When it is opposite, the rise of one feature uh you know, affects the you know, or rise of one feature uh brings down the other features' values, then we call it as negative covariance, okay? Then we have looked into how the covariance matrix is formed, what are the things that are present inside the matrix. We have seen the diagonal elements are variance of the of your components or the variables and the off-diagonal elements are the covariance between those you know, variables, okay? And we have finally we have taken up a numerical example where we have solved numerically for a very small data set having two features X_1 and X_2 . We have done the calculation for the covariance matrix, okay? So,

[31:22] with this one I come to the end of today's session. I hope you all had a understanding of the mathematical foundations that are required when we go ahead with our diffusion probabilistic models. In my next session, I will discuss about denoising diffusion probabilistic model, which is a specific case of diffusion probabilistic model and till that time keep learning, happy learning. Thank you all.